Artificial Intelligence

VICTORIA UNIVERSITY OF WELLINGTON TE HERENGA WAKA

AIML320/420

Decision Tree

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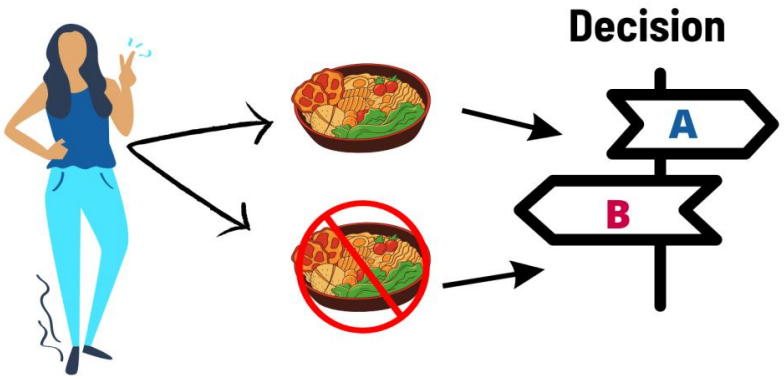
AIML320/420 Week 2

1. Announcements
   - Assignment 1 (**15%**)
     Due on 11:59pm,
     26th March (Thur), week 5
   - Helpdesk: Tuesday and Thursday, 2-3pm at CO242B
2. K nearest neighbour (KNN)
3. Decision Tree
   (able to work on Q1 of A1)

How do humans make decisions?

Adults: ~35000 decisions per day, ~2187.5 per hour, ~36.5 per minute, ~226.7 decisions each day on food alone



Have a think about the last decision you made…???

Brian Wansink & Jeffery Sobal (2007): Cornell University researchers, authors of "Mindless Eating: The 200 Daily Food Decisions We Overlook".
Sahakian & Labuzetta (2013): Mentioned in discussions regarding the 35,000 daily decision estimate.

How do humans make decisions?

Example: "Should I buy a car?"

What factors do you take into account?

1. *Lives far from work?*
2. *Is a Carpark available?*
3. *Afford it?*

A set of conditions!

- Organized in some **structures**
- In different **priority**

Decision tree techniques make this process into an algorithm;
Allows computers to automatically make decisions for us!

Outline

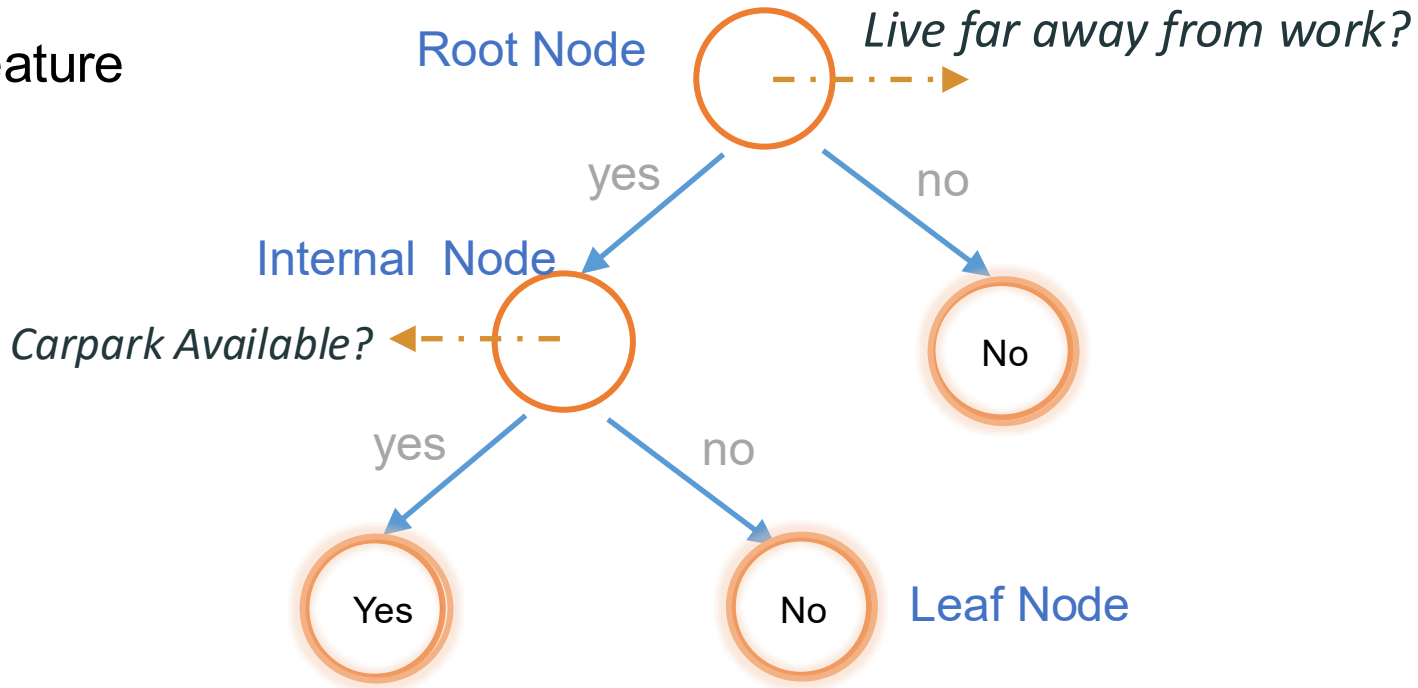
- What is a decision tree?
- How can we build decision trees?

What is a Decision Tree (DT)?

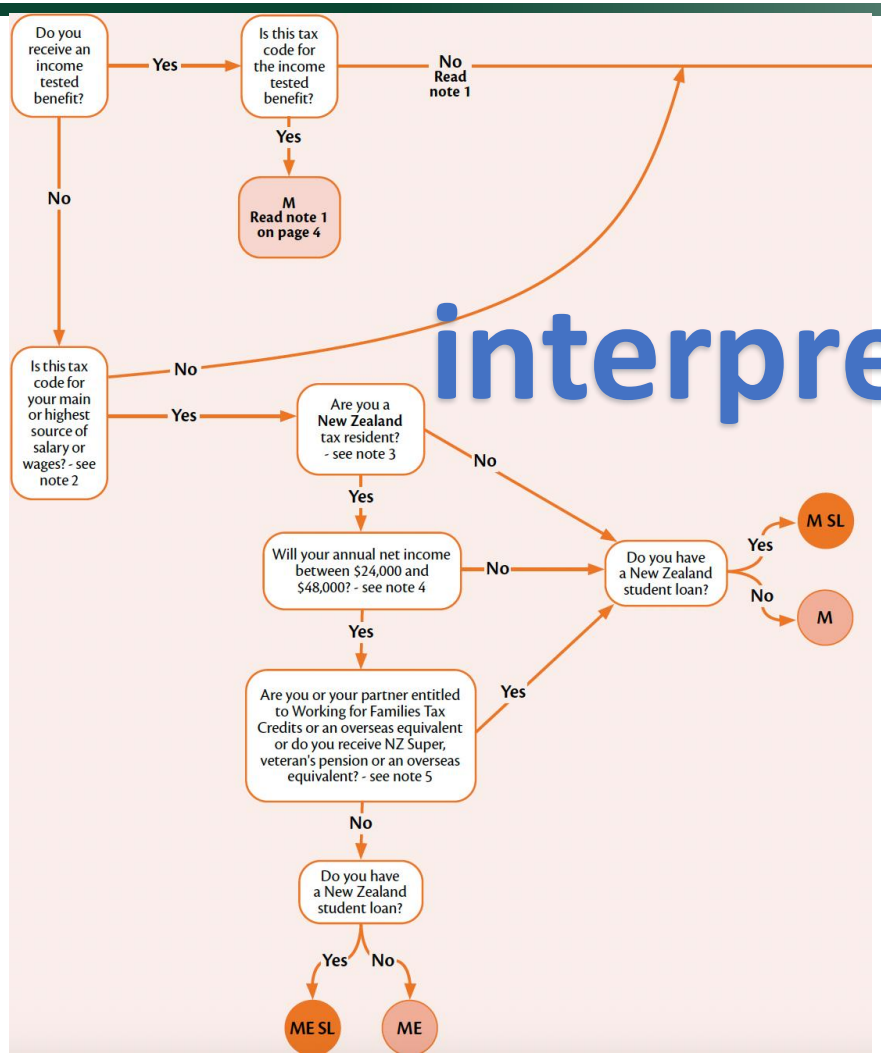
- **Decision Tree** is a tree-like machine learning model for making decisions

- Decision process is a path from root to leaves
- Each internal node tests an attribute/feature
- One branch for each possible attribute/feature value
- Each leaf node assigns a class y

Question: "Should I buy a car?" (Yes/No)



What is a Decision Tree (DT)?



interpretable

An Example of Decision Trees

Training Dataset

Day	Weather	Temp	Humid	Wind	Play
①	Sunny	Hot	High	Weak	No
②	Sunny	Hot	High	Strong	No
③	Rain	Hot	Low	Weak	Yes
④	Rain	Cool	High	Weak	No
⑤	Sunny	Cool	Low	Weak	Yes
⑥	Sunny	Hot	Low	Strong	No
⑦	Rain	Cool	Low	Strong	No
⋮	⋮	⋮	⋮	⋮	⋮

Question:

Play outside or not

Class:

Yes/ No

Attributes/Features/Conditions:

Weather: {Sunny, Rain}
Temperature: {Hot,Cool}
Humid: {High, Low}
Wind: {Strong, Weak}

Decision Trees Classification

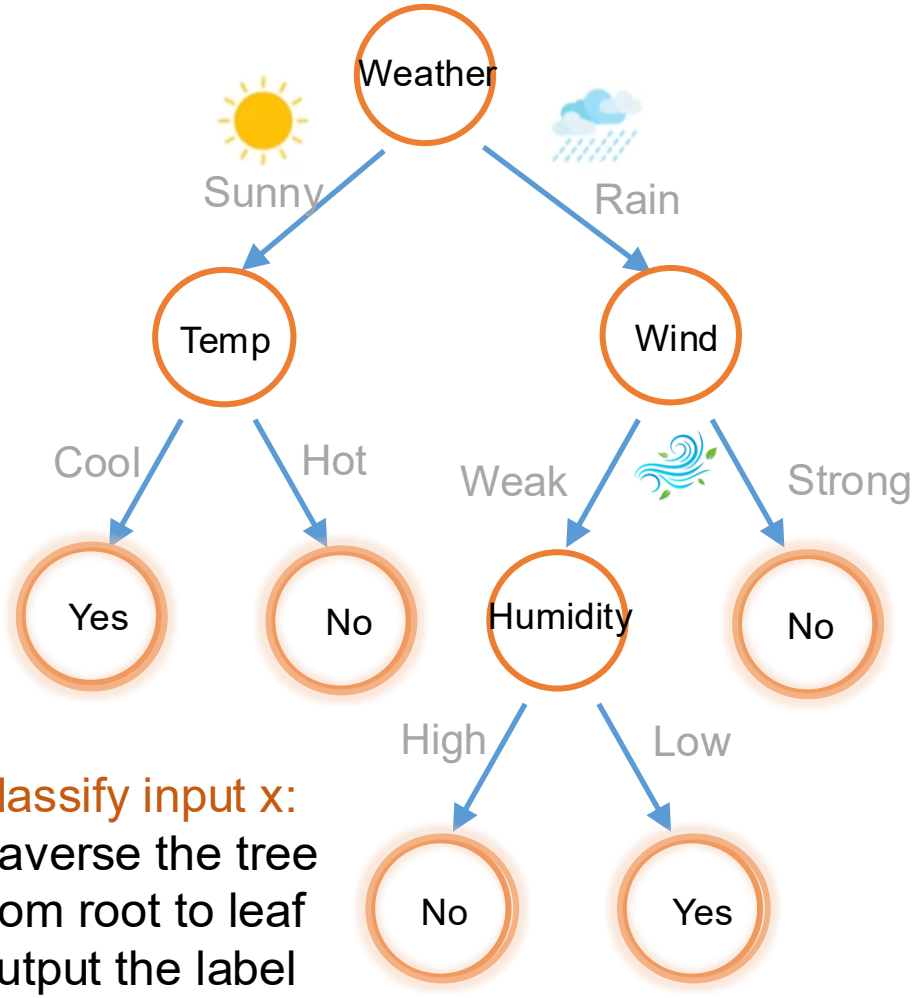
Training Dataset

Day	Weather	Temp	Humid	Wind	Play
①	Sunny	Hot	High	Weak	No
②	Sunny	Hot	High	Strong	No
③	Rain	Hot	Low	Weak	Yes
④	Rain	Cool	High	Weak	No
⑤	Sunny	Cool	Low	Weak	Yes
⑥	Sunny	Hot	Low	Strong	No
⑦	Rain	Cool	Low	Strong	No
⋮	⋮	⋮	⋮	⋮	⋮

Test Samples

Day	Weather	Temp	Humid	Wind	Play
①	Sunny	Cool	High	Weak	?
②	Rain	Hot	High	Weak	?

$y = f(x)$



To classify input x:

- traverse the tree from root to leaf
- output the label of the leaf finally reached

Yes

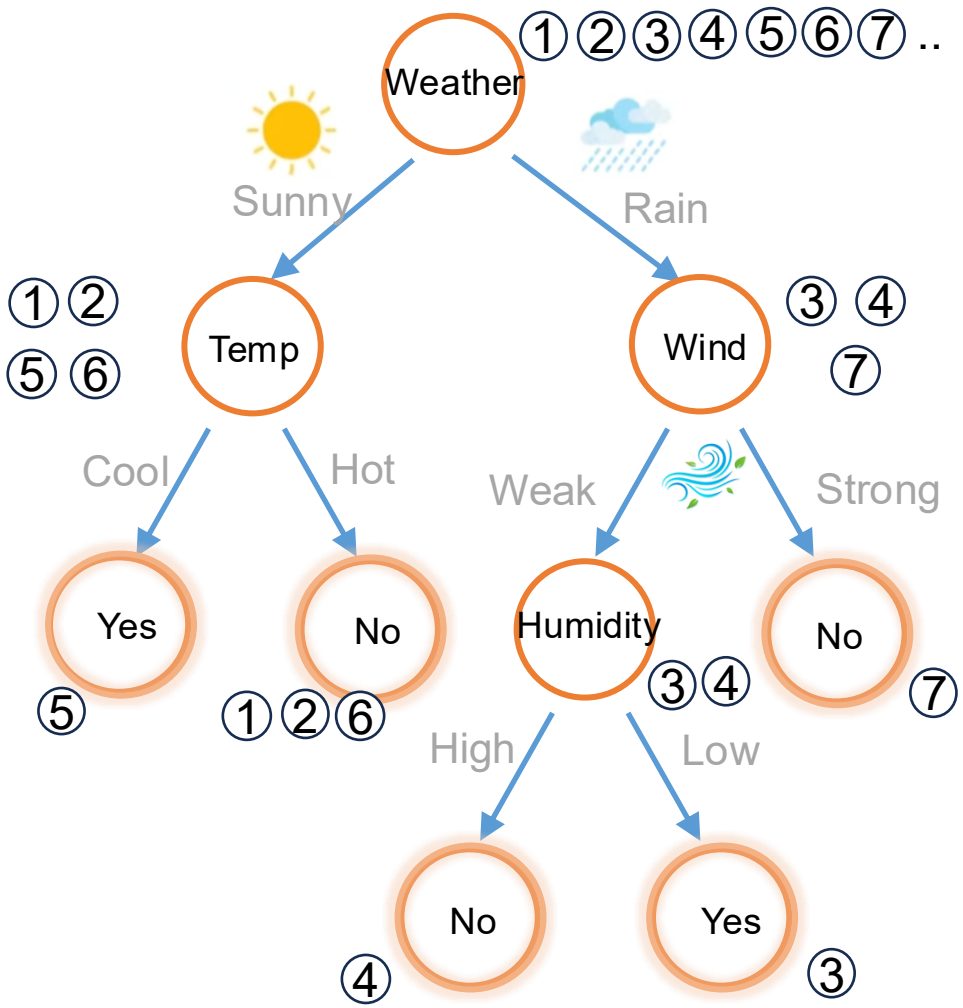
No

Build a Decision Tree

Input: Instances set
General Process: A tree-building process is a data-splitting process

Training Dataset

Day	Weather	Temp	Humid	Wind	Play
①	Sunny	Hot	High	Weak	No
②	Sunny	Hot	High	Strong	No
③	Rain	Hot	Low	Weak	Yes
④	Rain	Cool	High	Weak	No
⑤	Sunny	Cool	Low	Weak	Yes
⑥	Sunny	Hot	Low	Strong	No
⑦	Rain	Cool	Low	Strong	No
⋮	⋮	⋮	⋮	⋮	⋮



Build a Decision Tree

Input: Instances set

Output: A good decision tree classifier

- Start: Build a tree with a single root node
- Repeat until no split is needed:
  - Examine the set of instances in the node

    If the instance set is with the same label (pure):

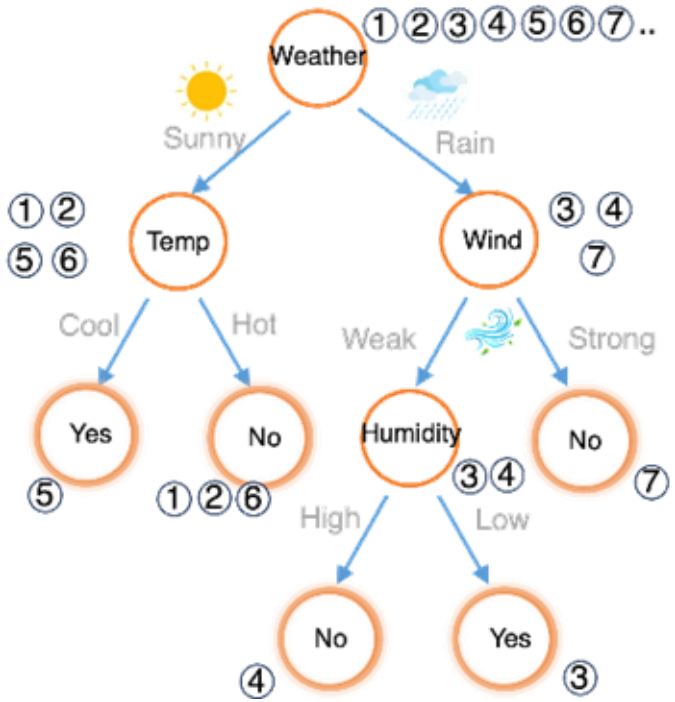
    Take the node as a leaf node

    else:

    Select the best attribute to split the set for the child nodes

    Build child nodes repeatly with the subsets of instances

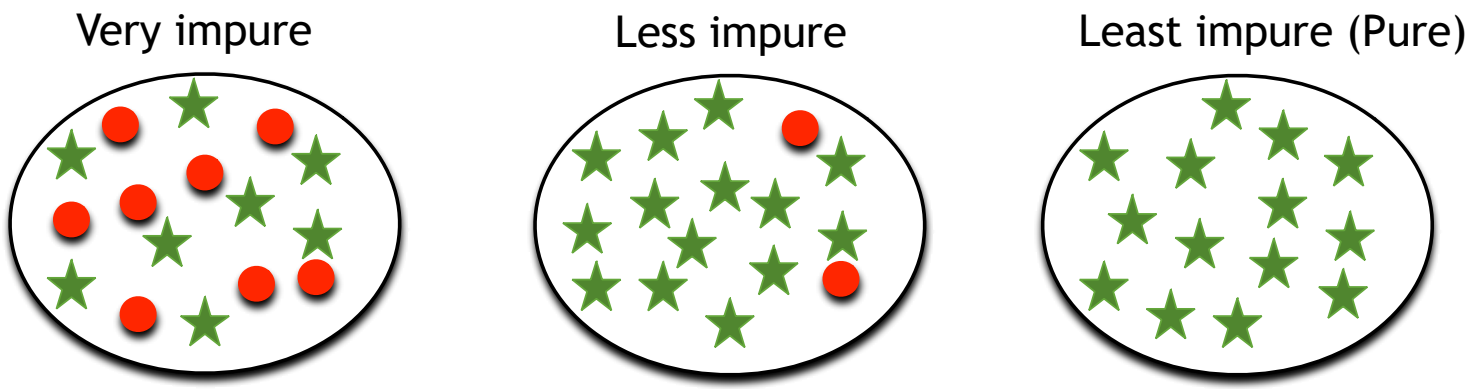
Use a criteria to control the data splitting!



It's greedy because every time select the best one

What is "Best Attribute"? Impurity Measurement

Pure or Impure?



- Impure->Non-Deterministic: <u>bad</u>
  - mixed classes in the node
  - need to ask more questions
- Pure->Deterministic: <u>good</u>
  - just one class in the node
  - can predict a class

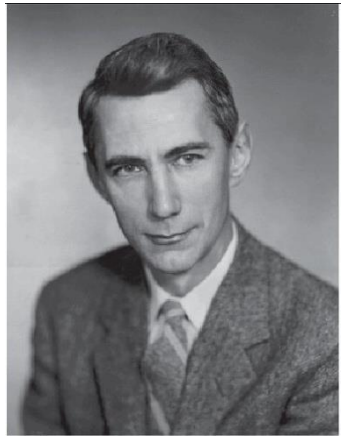
Best Attribute: The splitting should make the child nodes as pure as possible

Node (im)purity: can be defined in different ways

- Probability-based
- <u>Information theory based, i.e., </u>Information Gain <--- Entropy

Quantifying Impurity: Entropy

Entropy :



Claude Shannon (1916-2001)

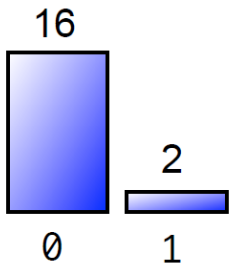
$$H(X) = -\sum_{x \in X} p(x) \log_2 p(x)$$

It indicates how much **uncertainty** is present in a group of samples.

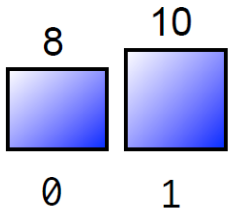
P(X): Probability of X

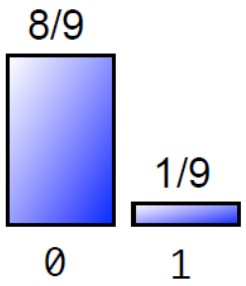
Sequence 1:
0 0 0 1 0 0 0 0
0 0 0 0 0 0 0 1
0 0

Sequence 2:
0 1 0 1 0 1 1 1
0 1 0 0 1 1 0 1
0 1

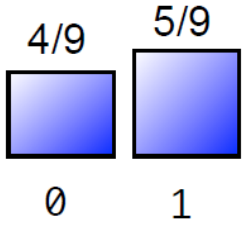


versus





versus



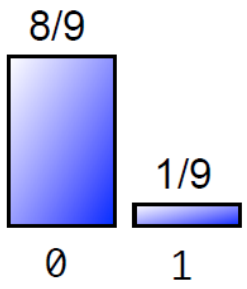
$$-\frac{8}{9}\log_2\frac{8}{9} - \frac{1}{9}\log_2\frac{1}{9} \approx \frac{1}{2}$$

$$-\frac{4}{9}\log_2\frac{4}{9} - \frac{5}{9}\log_2\frac{5}{9} \approx 0.99$$

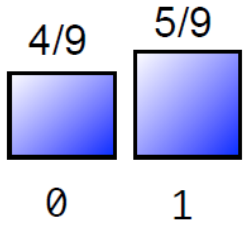
C.E. Shannon, 1948, "A mathematical theory of communication", Bell System Technical Journal, Vol. 27, pp.379-423 and 623-656 (1948)

Quantifying Impurity: Entropy

- Entropy: $$\mathrm{H(X)} = -\sum_{i=1}^{C} P(\mathrm{i})\log_2 P(i)$$



$$-\frac{8}{9}\log_2\frac{8}{9} - \frac{1}{9}\log_2\frac{1}{9} \approx \frac{1}{2}$$



$$-\frac{4}{9}\log_2\frac{4}{9} - \frac{5}{9}\log_2\frac{5}{9} \approx 0.99$$

Entropy measures the uncertainty over the value of a random variable

<u>The smaller, the better!</u>

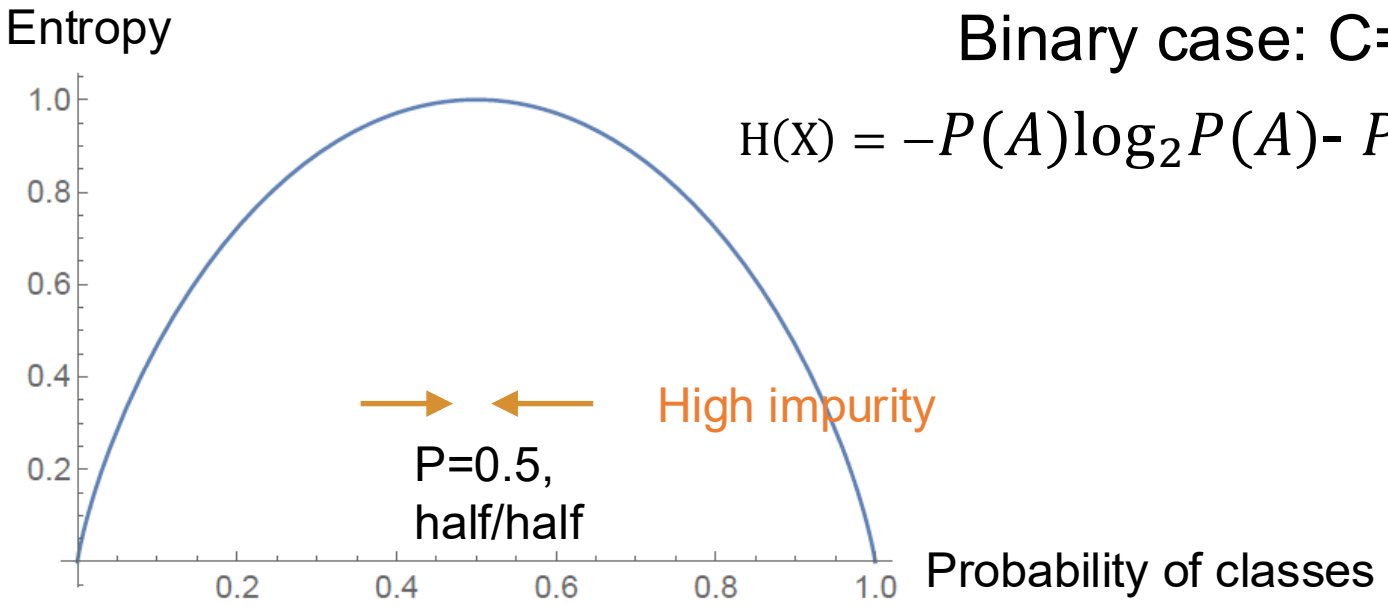
Low Entropy:
- Distribution of variable has many peaks and valleys
- Histogram has many lows and highs
- More predictable

High Entropy:
- Variables have a uniform like distribution
- Flat histogram
- Less predictable

Quantifying Impurity: Entropy

Entropy : $\mathrm{H(X)} = -\sum_{i=1}^{C} P(\mathrm{i})\log_2 P(i)$

Smooth



Binary case: C=2

$\mathrm{H(X)} = -P(A)\log_2 P(A)\text{-}\ P(B)\log_2 P(B)$

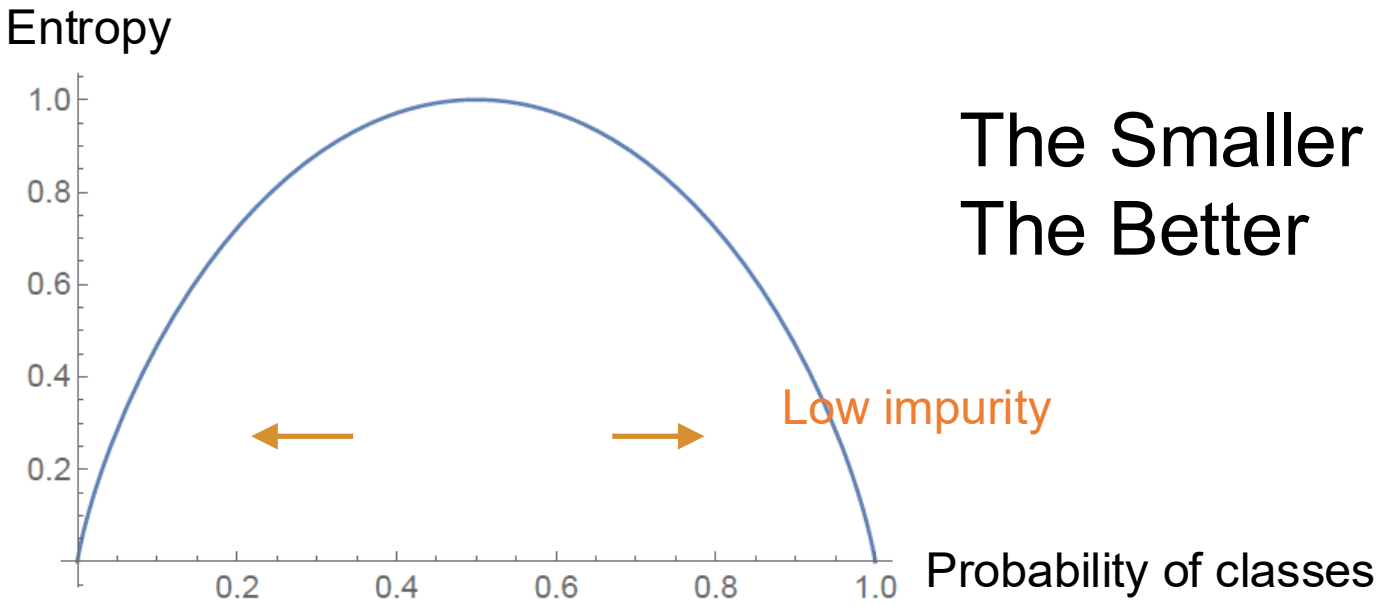
Quantifying Impurity: Entropy

Entropy : $\mathrm{H(X)} = -\sum_{i=1}^{C} P(\mathrm{i})\log_2 P(i)$

Binary case: C=2

$\mathrm{H(X)} = -P(A)\log_2 P(A) - P(B)\log_2 P(B)$

Smooth



The Smaller
The Better

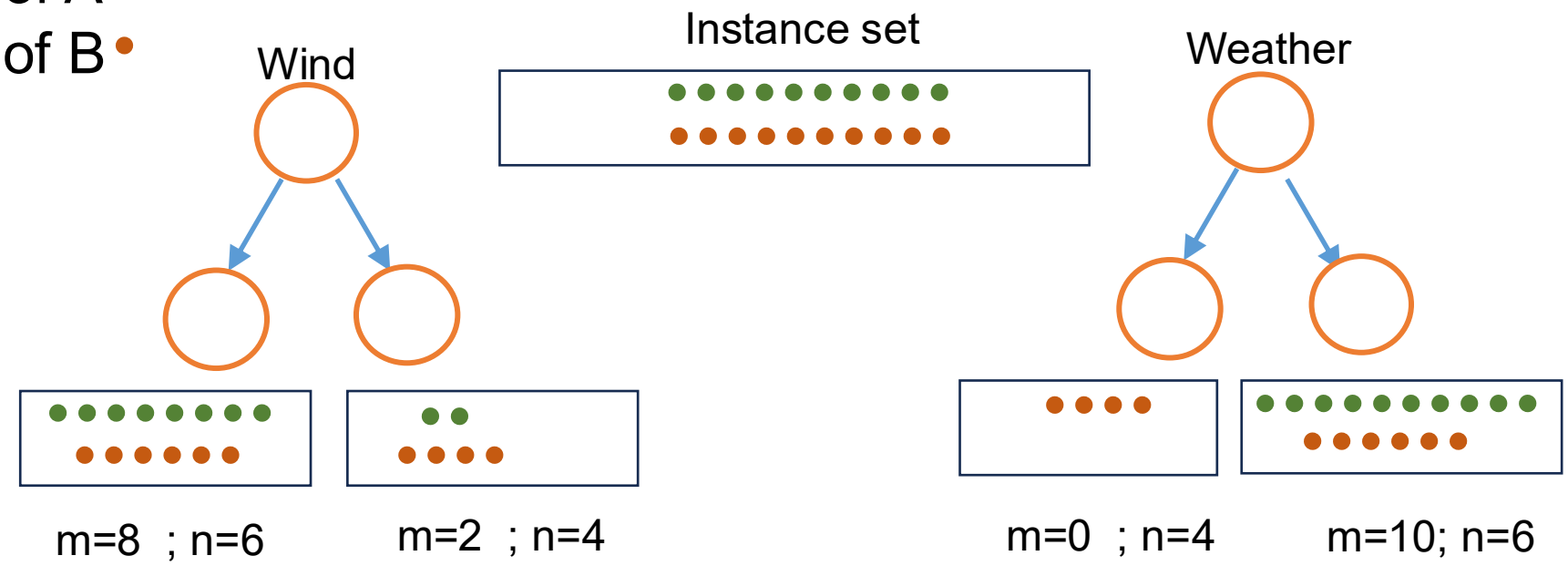
Choose a Good Attribute: Impurity of Nodes

$$H(X) = -P(A)\log_2 P(A) - P(B)\log_2 P(B)$$

Which attribute is better for splitting?

m: number of A

n: number of B



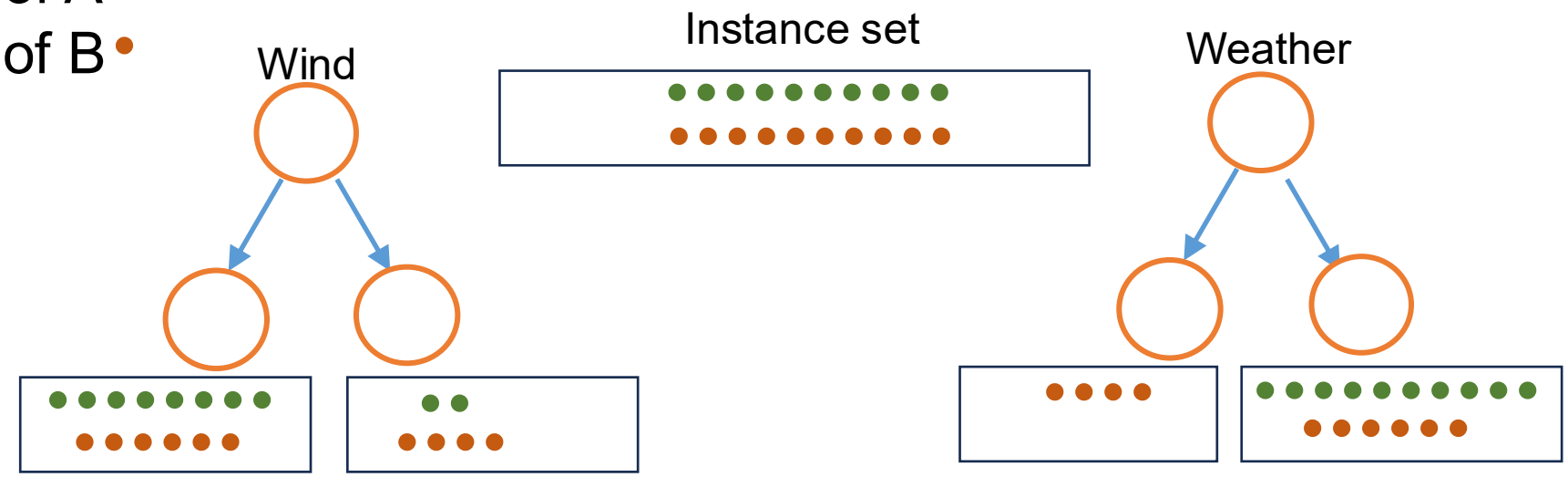
Choose a Good Attribute: Impurity of Nodes

$\mathrm{H(X)} = -P(A)\log_2 P(A) - P(B)\log_2 P(B)$

Which attribute is better for splitting?

m: number of A ●

n: number of B ●



m=8 ; n=6

$$p(A) = \frac{8}{8+6} \approx 0.57$$

$$p(B) = \frac{6}{8+6} \approx 0.43$$

m=2 ; n=4

$$p(A) = \frac{2}{2+4} \approx 0.33$$

$$p(B) = \frac{4}{2+4} \approx 0.67$$

m=0 ; n=4

$$p(A) = \frac{0}{0+4} = 0$$

$$p(B) = \frac{4}{0+4} = 1$$

Pure node

m=10: n=6

$$p(A) = \frac{10}{10+6} \approx 0.63$$

$$p(B) = \frac{6}{10+6} \approx 0.38$$

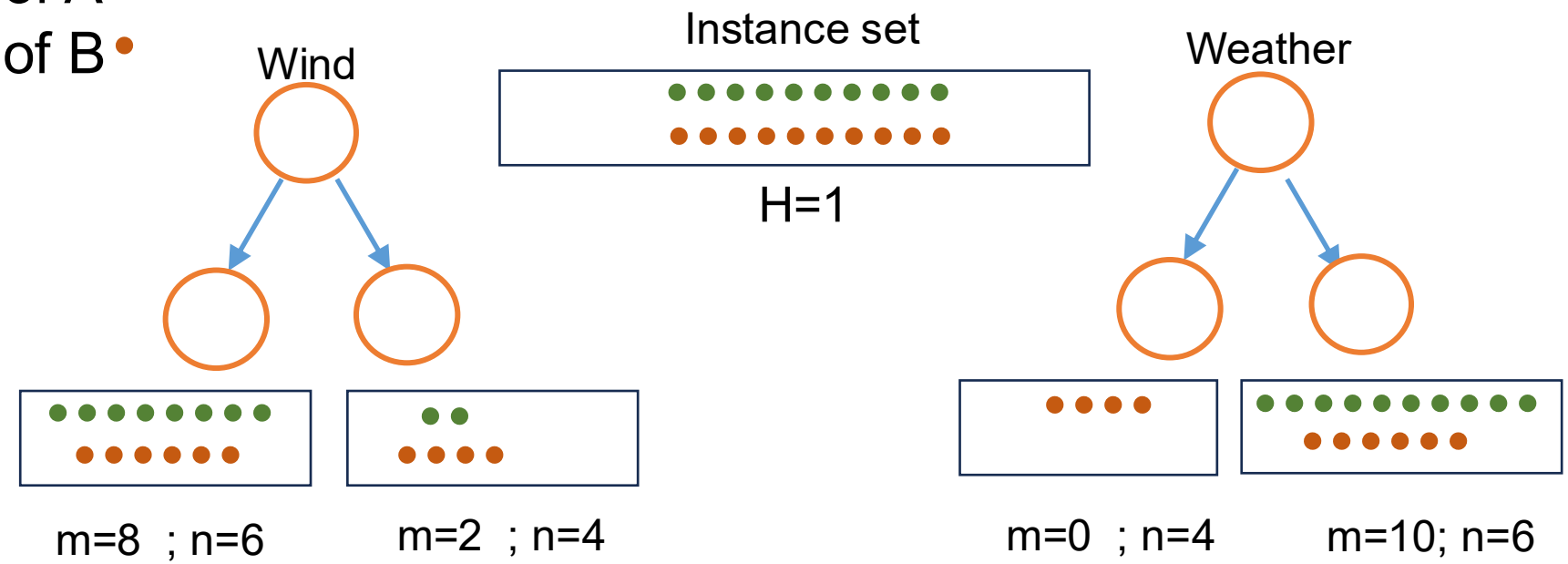
Choose a Good Attribute: Impurity of Nodes

$H(X) = -P(A)\log_2 P(A) - P(B)\log_2 P(B)$

Which attribute is better for splitting?

m: number of A

n: number of B



$$p(A) = \frac{8}{8+6} \approx 0.57$$

$$p(B) = \frac{6}{8+6} \approx 0.43$$

$$H(X) = -P(A)log_2 P(A) - P(B)log_2 P(B)$$

$$= -0.57 * log_2(0.57) - 0.43 * log_2(0.43)$$

=0.99

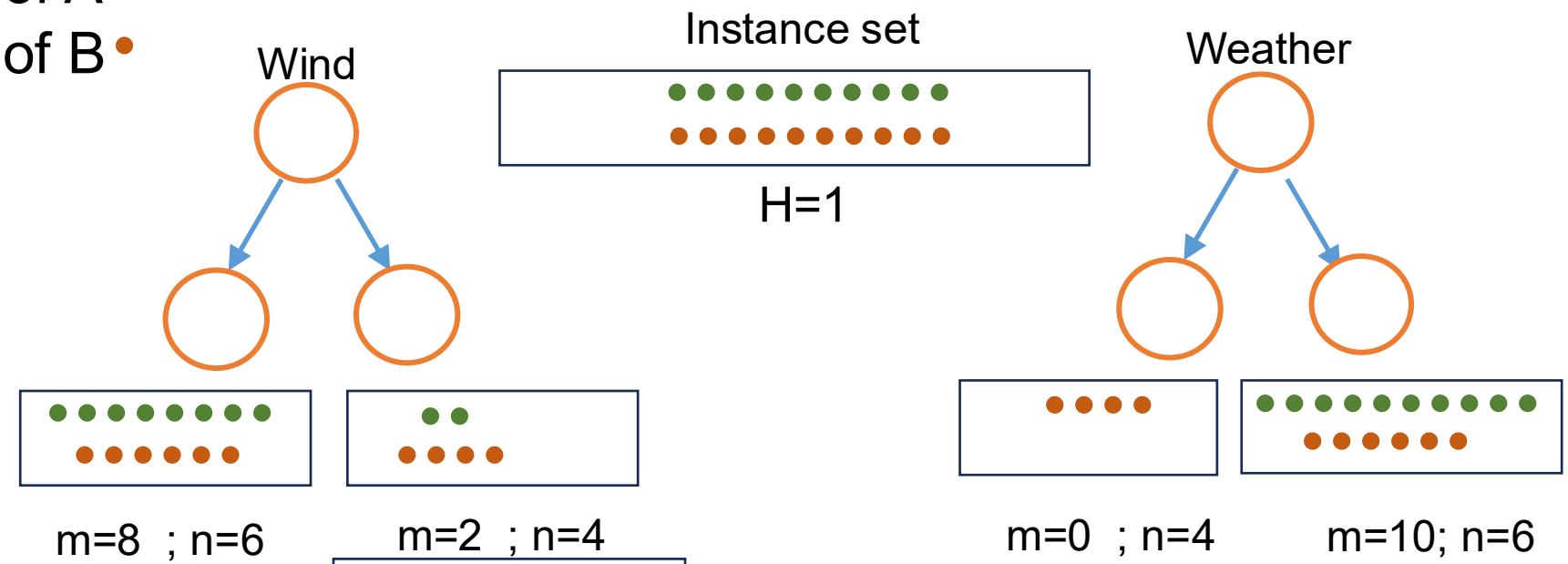
Choose a Good Attribute: Impurity of Nodes

$\mathrm{H(X)} = -P(A)\log_2 P(A) - P(B)\log_2 P(B)$

Which attribute is better for splitting?

m: number of A

n: number of B



$$p(A) = \frac{2}{2+4} \approx 0.33$$

$$p(B) = \frac{4}{2+4} \approx 0.67$$

$$\mathrm{H(X)} = -P(A)\log_2 P(A) - P(B)\log_2 P(B)$$
$$= -0.33 * \log_2 (0.33) - 0.67 * \log_2 (0.67)$$
$$= 0.91$$

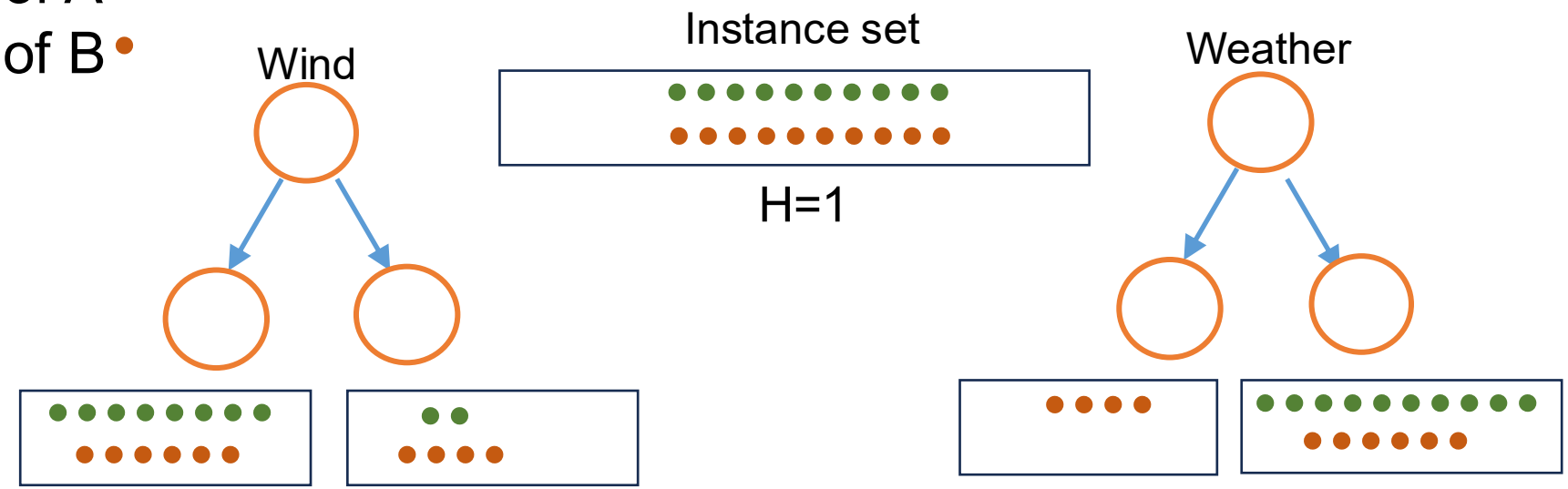
Choose a Good Attribute: Impurity of Nodes

$H(X) = -P(A)\log_2 P(A) - P(B)\log_2 P(B)$

Which attribute is better for splitting?

m: number of A •

n: number of B •



$$p(A) = \frac{0}{0+4} = 0$$

$$p(B) = \frac{4}{0+4} = 1$$

$H(X) = -P(A)\log_2 P(A) - P(B)\log_2 P(B)$

$= -0 * \log_2(0) - 1.0 * \log_2(1.0)$

=0

Pure node entropy is 0

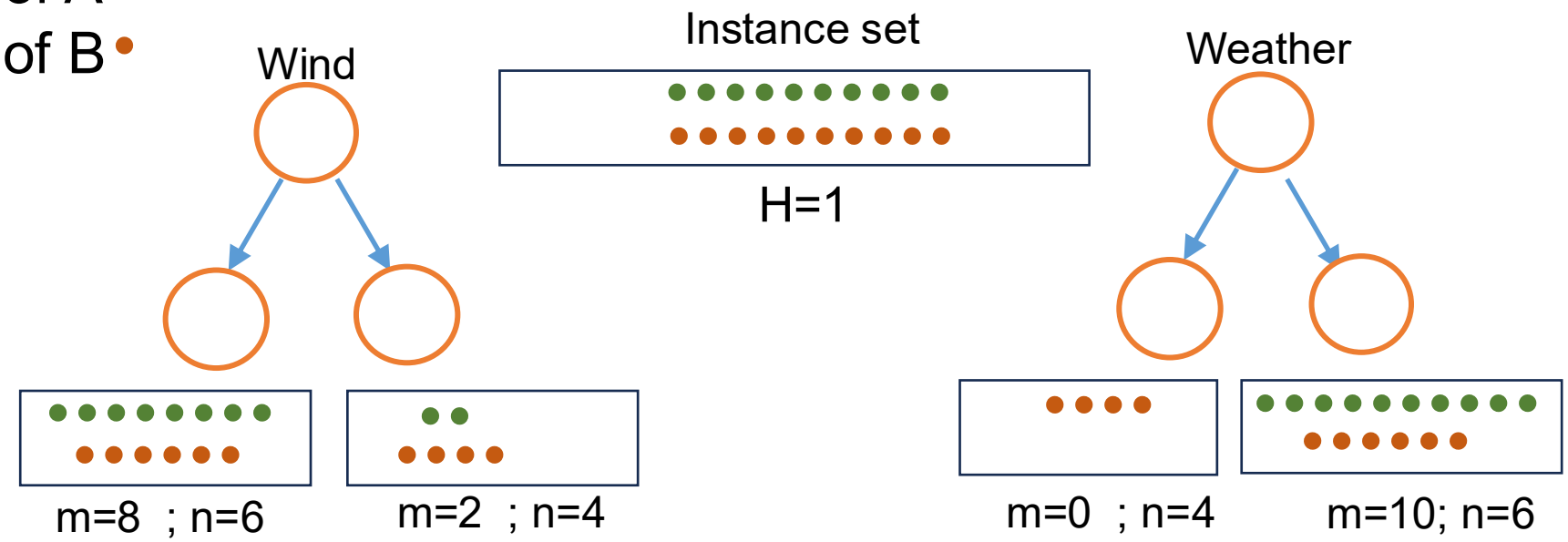
Choose a Good Attribute: Impurity of Nodes

$\mathrm{H(X)} = -P(A)\log_2 P(A) - P(B)\log_2 P(B)$

Which attribute is better for splitting?

m: number of A

n: number of B



$$\mathrm{H(X)} = -P(A)\log_2 P(A) - P(B)\log_2 P(B)$$

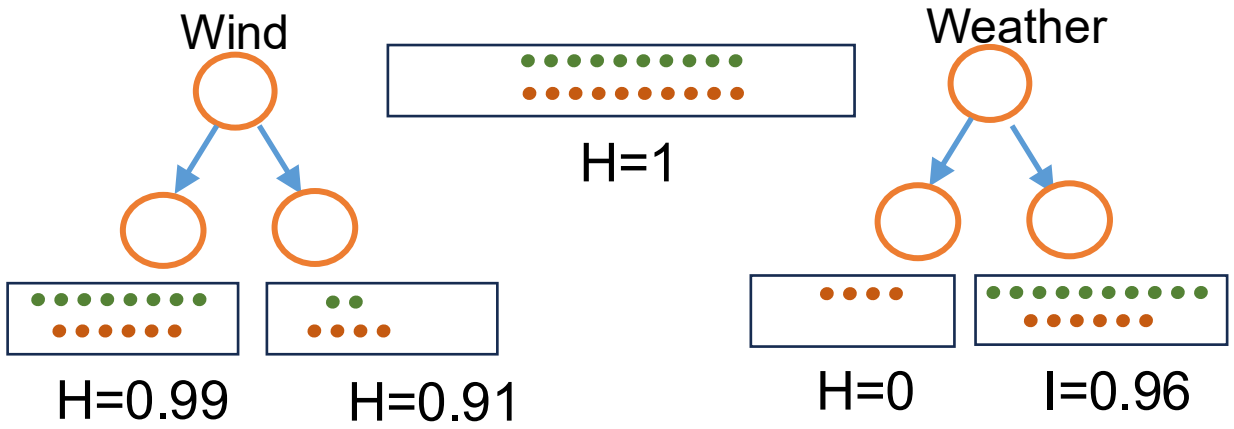
$$= -0.63 * \log_2(0.63) - 0.38 * \log_2(0.38)$$

=0.96

$$p(A) = \frac{10}{10+6} \approx 0.63$$

$$p(B) = \frac{6}{10+6} \approx 0.38$$

Choose a Good Attribute



- Goodness of an attribute: Compare the entropy before and after splitting
- **Information Gain (IG):** how much entropy will be reduced by splitting data on a particular attribute

Entropy → Information Gain (IG)

Choose a Good Attribute - Information Gain (IG)

$$IG = H(parent) - \sum w_i H(child)$$

H(parent): entropy of parent's node

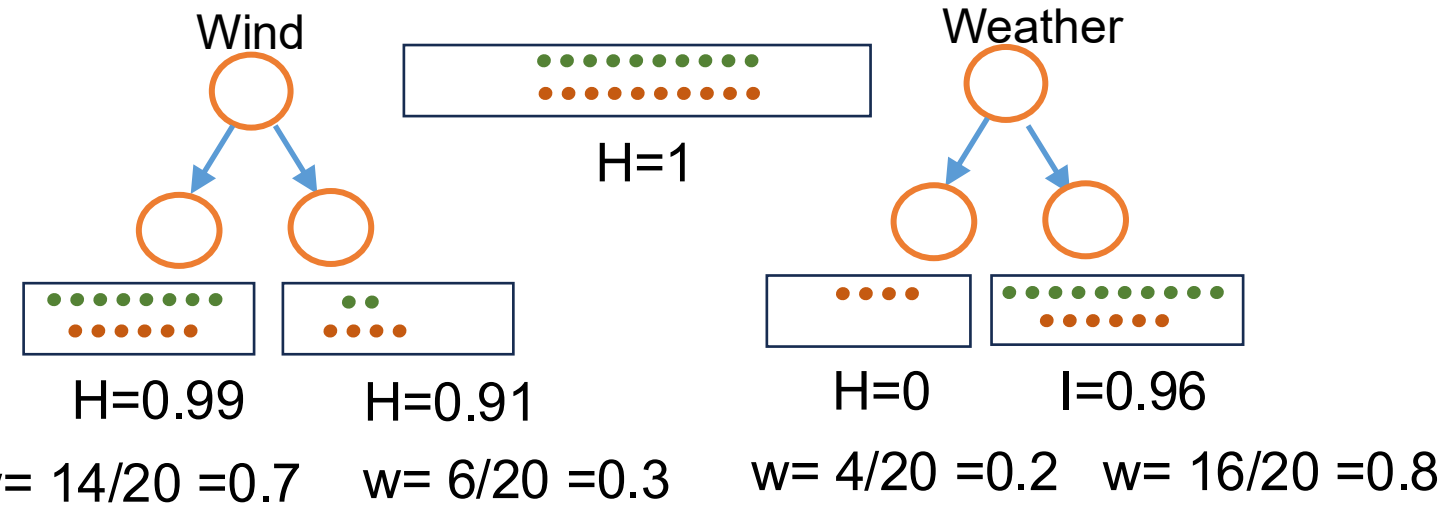
H(child): entropy of child node

$$w(i) = \frac{\#instances\ in\ node\ i}{\#instances\ in\ node\ i's\ parents}$$

IG calculation:

- Calculate the entropy of parent node
- Calculate the weighted entropy of all child nodes (>=2)
- Calculate the difference between parent and child nodes to get the IG
- Choose the splits with higher IG

Choose a Good Attribute



- Goodness of an attribute: Compare the entropy before and after splitting:
- Information Gain (IG): $IG = H(parent) - \sum w_i H(child)$

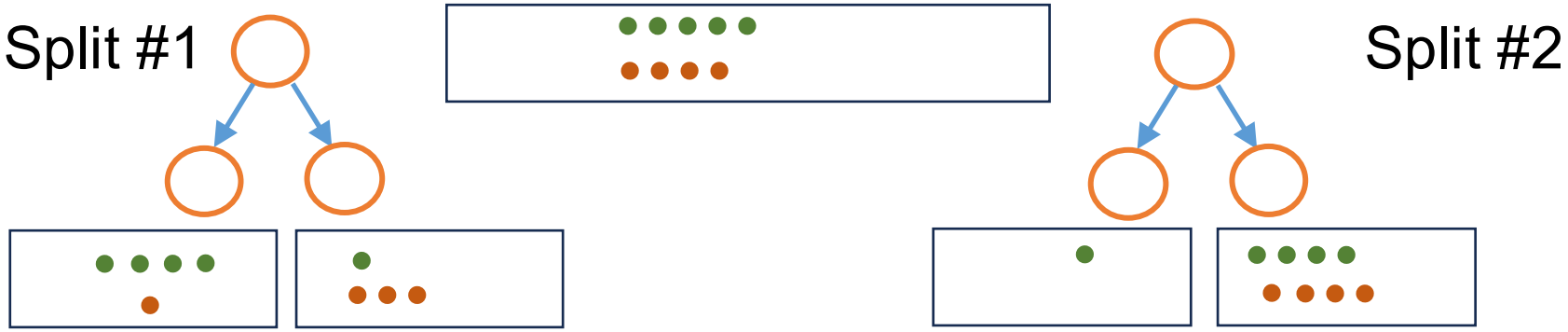
$$IG(wind) = 1 - (0.7 * 0.99 + 0.3 * 0.91) = 1 - 0.966 = 0.034$$

$$IG(weather) = 1 - (0.2 * 0 + 0.8 * 0.96) = 1 - 0.768 = 0.232$$

$$IG(wind) < IG(weather)$$

Choose "Weather"

Why weighted Average? Another Example



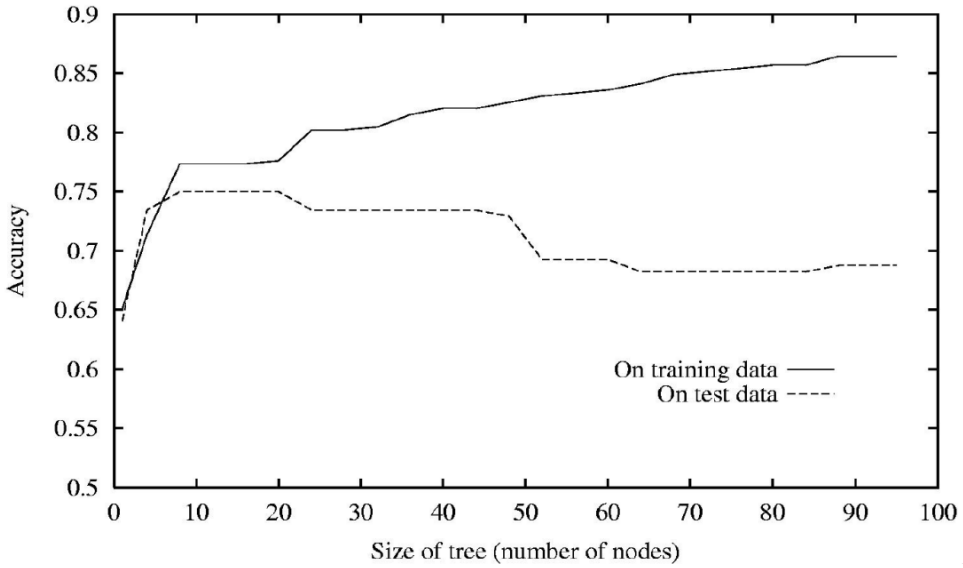
- The normal average can be biased to uneven splits (which one branch has very less instance).
- Splits with less instance tend to make tree too deep, i.e., Decision Tree begins to "memorize" the noise of the training data rather than learning general patterns.

When to Stop Splitting? – Depth Control

- If stop splitting too early, then the node is not pure enough
- If stop splitting too late, then the tree becomes too large and complex, and can be overfitting
  - The accuracy on the training set can be high. However, the accuracy on the test set can be poor.

Stop splitting a node when

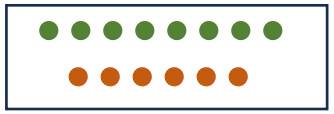
- A node is pure (or reach some impurity threshold)
- The maximal tree depth/size is reached
- The best candidate split reduces the impurity by less than the preset threshold (e.g. 5%)
- …



Other Impurity Functions: Gini Impurity

- Gini impurity: $\mathrm{Gini(X)} = \sum_{i=1}^{C} P(i) \sum_{j \neq i} P(j) = 1 - \sum_{i=1}^{C} P(i)^2$

  Where P(i) is the proportion of instances in the nodes that belong to class i



m=8 ; n=6

$$p(A) = \frac{8}{8+6} \approx 0.57$$

$$p(B) = \frac{6}{8+6} \approx 0.43$$

$$Gini(x) = 1 - P^2(A) - P^2(B) = 1 - 0.57^2 - 0.43^2 = 0.49$$

Various Decision Tree Learning Algorithms

- Iterative Dichotomiser 3 (ID3)
  - Information Gain based on entropy
- C4.5/ C5.0
  - Gain ratio
- Classification and Regression Tree (CART)
  - Gini impurity, a binary tree
- And others…

The principle is just what we talked about!

Summary

- What's a Decision Tree
- How to build a decision
  - Algorithm ---- greedy approach
  - Impurity measure: several methods, particularly entropy and GI
  - Stopping criteria
- Provide template for Q1 of A1, check course website

Next:

- Guest Lecture: History of AI